Do Now · retrieval

Lesson 079 · Bubble Sort: Adjacent Swaps and Passes

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which elements does bubble sort compare?

Q2

What is placed by one ascending pass?

Q3

What does swap do?

Q4

Swap elements at indexes 0 and 1 in array [4,2,6]. Give the new array.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Compare adjacent pairs, swapping them when out of ascending order.

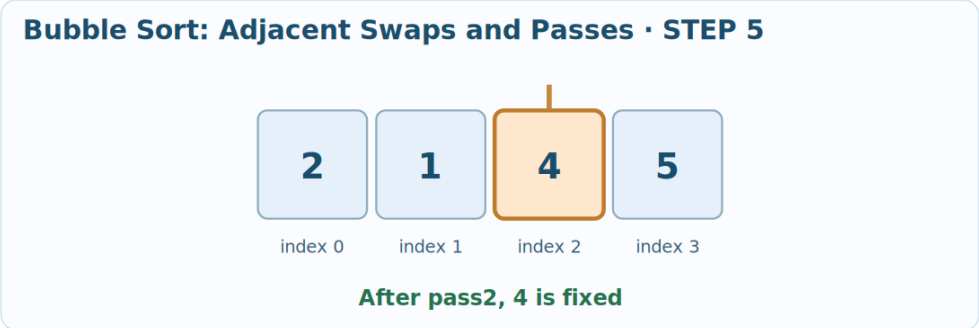
Core idea 2

Trace — One completed ascending pass places the largest remaining element at the right end of the unsorted prefix.

Core idea 3

Explain — Straightforward bubble sort is O(n²) worst time and O(1) extra memory; an early-exit flag can give O(n) best time.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

bubble sort — Compare/swap adjacent values.

pass — Sweep through neighbouring pairs.

swap — Exchange two elements.

invariant — Always true state after given stage.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

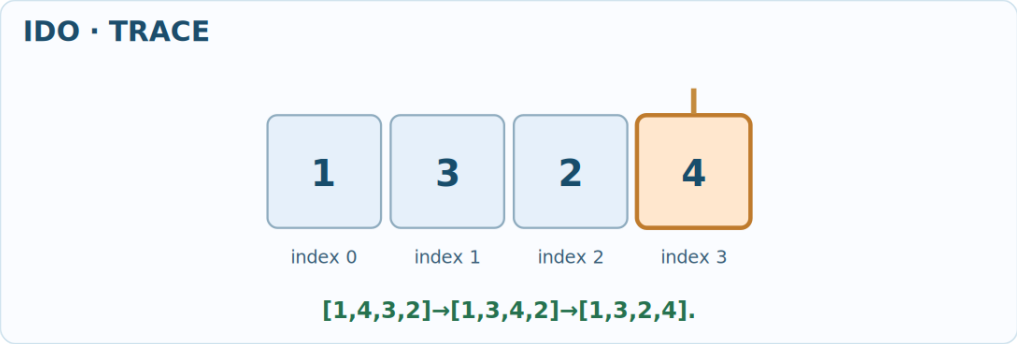
Problem

Show full first pass [4,1,3,2].

Follow the worked method

1. Check the working — [1,4,3,2]→[1,3,4,2]→[1,3,2,4].

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Check the working
[2,3,5,1] (swap); 3,5 no swap; [2,3,1,5] (swap).

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give all states from first pass [5,2,4,1]. [6]

Question 2

Give pass-end arrays for [5,2,4,1] until sorted. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain bubble-sort worst-time and extra-space complexity and benefit of early-stop. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

After ascending first bubble pass, smallest value must be fixed at index0.

Use key vocabulary from memory

1. Explain bubble sort in your own words.

2. Explain pass in your own words.

3. Explain swap in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.